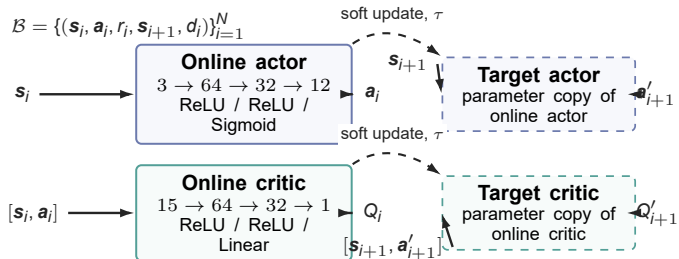


Network architecture



Learning equations

1. TD target

$$y_i = r_i + \gamma(1 - d_i)Q'(\mathbf{s}_{i+1}, \mu'(\mathbf{s}_{i+1}))$$

2. Critic loss

$$L_Q = \frac{1}{N} \sum_i [Q(\mathbf{s}_i, \mathbf{a}_i) - y_i]^2$$

3. Actor objective

$$L_\mu = -\frac{1}{N} \sum_i Q(\mathbf{s}_i, \mu(\mathbf{s}_i))$$

4. Soft updates

$$\theta^{\mu'} \leftarrow \tau \theta^\mu + (1 - \tau) \theta^{\mu'}$$

$$\theta^{Q'} \leftarrow \tau \theta^Q + (1 - \tau) \theta^{Q'}$$

$$N = 128, \quad \gamma = 0.999, \quad \tau = 0.001$$